

Title Estimating duration in sex work from cross-sectional studies

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Code github.com/mishra-lab/sw-duration

Abstract

Keywords

1 Introduction

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Objectives Our objectives were to:

1. Describe the analytic relationship between the distribution of reported durations selling sex in observational studies, and the distribution of total durations selling sex in the population overall.
2. Develop and apply a Bayesian model which leverages the above relationships to infer the distribution of total durations selling sex in the population overall from published data.

While we focus on sex work as an illustrative example, our insights extend to any transient risk/disease state, relevant to compartmental modelling or epidemiological study. Indeed, similar work has previously examined estimation of disease state "sojourn times" in models of chronic diseases [1].

2 Preliminaries

Assumptions For simplicity, we first assume that periods of sex work can be precisely and instantaneously delineated, i.e. overlooking blurry boundaries between sex work and transactional sex, and definitions of sex work based on activity in the past 6 months, etc. [2]. We further assume that survey participants can perfectly recall periods of sex work. Finally, we assume that all populations are at equilibrium, despite empiric evidence of changing sex work dynamics in several contexts [3].

Existing estimation methods In their global scoping review of HIV risk behaviour durations, Fazito et al. [4] considered three methods to estimate mean duration selling/buying sex (Table 1): 1. mean total duration among those finished selling/buying sex; 2. reciprocal of the proportion who started selling/buying sex within the past year; and 3. double the mean/median reported duration among those actively selling/buying sex. Notably, among the 118 estimates of duration selling or buying sex identified by Fazito et al.: the vast majority used method 3 (86%), a small fraction used method 2 (14%), and none used method 1. Studies also commonly report the proportions of respondents selling/buying sex for pre-specified duration ranges, e.g. {0–1, 1–5, 5+} years, though it's not clear if/how such data were used in [4].

Who exactly are we talking about? Before continuing, it is helpful to distinguish two populations: individuals who will ever sell/buy sex in their lifetime, and individuals who are actively selling/buying sex. The former "source population" is the sampling frame for method 1, and tends to include more individuals who sell/buy sex for shorter durations; the latter "active population" is the sampling frame for methods 2 and 3, and tends to include more individuals who sell/buy sex for longer durations (Figure 1). It is tempting to think that, since "sex work" strata in compartmental models reflect the active population, rates of exit from sex work should be derived from this population. However, this is not the case, because these rates apply to all individuals *entering* sex work, which reflects the source population.¹ A final population worth distinguishing is the "sampled population" in a given cross-sectional survey, which we assume intends to be representative of the active population, but might undersample some individuals, e.g. those in the early years of sex work, due to delays in self-identification as a sex worker [5, 6].

¹ An analogous example: the mean duration of an infection should be characterized by a cohort of individuals with *incident* infections, not a cross-section of individuals with *prevalent* infections.

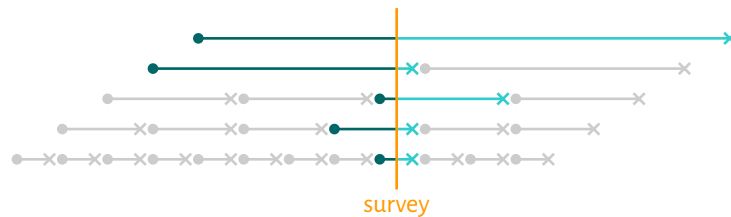


Figure 1: Diagram of periods selling/buying sex in an open population at equilibrium

Colours — dark green: past sex work among active population (observed); light green: future sex work among active population (unobserved); gray: past/future total sex work periods among inactive population (unobserved).

Table 1: Methods of estimating mean duration selling sex suggested and reviewed in [4]

	Definition	Potential challenges	N [†]
1.	mean total duration among those finished selling sex	• very difficult to sample this population	0
2.	reciprocal of the proportion who started selling sex within the past year among those actively selling sex	• this proportion is rarely reported • sensitive to delays in self-identification • not adjusted for sampling bias	16
3.	double the mean/median reported (right-censored) duration among those actively selling sex	• median < mean for skewed distributions • not adjusted for sampling bias	93

[†] Number of estimates of duration selling sex synthesized by [4]; corresponding numbers were: 0, 1, 8 for duration buying sex.

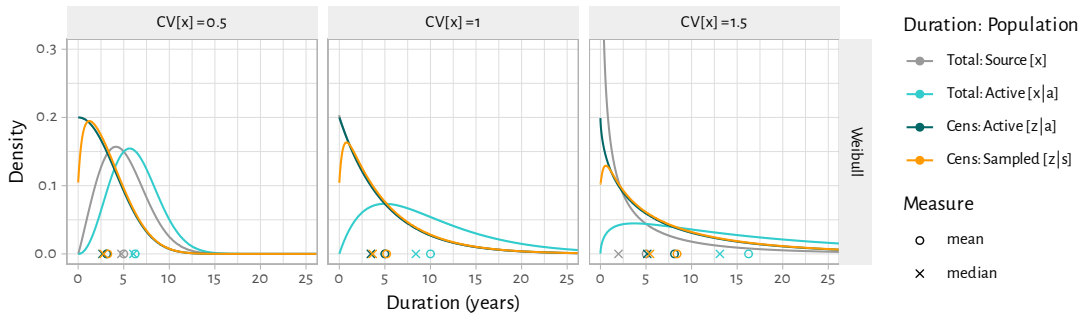


Figure 2: Illustrative distributions of total and censored durations among the source, active, and sampled populations for 3 coefficients of variation $CV[x]$ (rows)

Assumed: $E[x] = 5$ and $x \sim \text{Weibull}$ (Figure A.5 illustrates other distribution families); probability of being sampled per Eq. (7).

3 Analytic relationships

We now turn to deriving the exact relationship between: the distribution of *total* durations selling sex among the source population: $x \sim p(x)$, and the distribution of *observed* durations selling sex among participants in a cross-sectional survey: $z \sim p(z|s)$, given survey participation status s . We also consider two survey sampling cases: an ideal case which is representative of the active population, and a more realistic case which undersamples individuals in the early years of sex work. For reference throughout this section, Figure 2 illustrates example distributions for $p(x)$, $p(z|s)$, and intermediates thereof.

Active population bias We begin by reiterating that individuals who sell sex for longer are more likely to be active at any given time. Specifically, the probability of an individual being active is proportional to their total duration x . Thus, the distribution of total durations among the active population is:

$$p(x|a) = \frac{1}{K} x p(x) \quad (1)$$

where K is a normalizing constant. These total durations among the active population are typically unobserved, except in cohort studies which follow participants until they exit sex work. From Eq. (1) we can see that the

mean total duration among the active population $\mathbb{E}[x|a]$ is always larger than the mean among the source population $\mathbb{E}[x]$ (Figure A.5, light green vs gray circles); specifically, we have that:

$$\mathbb{E}[x|a] = \mathbb{E}[x] (1 + \mathbb{CV}^2[x]) \quad (2)$$

This is because greater variability in durations among the source population, as measured by the coefficient of variation: $\mathbb{CV}[x]$, allows those with longer durations to increasingly dominate the active population. We also note that the probability of observing active individuals with very short total durations is inherently very small (Figure 2, light green).

Censoring If survey timing is effectively random, then observed durations at the time of survey (z) will be right-censored and uniformly distributed between 0 and the total duration (x):

$$p(z|x) = \text{Unif}(0, x) = \begin{cases} \frac{1}{x} & \text{if } 0 \leq z \leq x \\ 0 & \text{else} \end{cases} \quad (3)$$

Since the mean of $\text{Unif}(0, x)$ is $\frac{x}{2}$, the mean censored duration is always half the mean total duration among the active population: $\mathbb{E}[z|a] = \frac{1}{2} \mathbb{E}[x|a]$ (Figure 2, dark green vs light green circles). This idea appears to be well-known, and likely motivates doubling the mean $\mathbb{E}[z|a]$ to estimate $\mathbb{E}[x]$ in method 3 of [4]. However, method 3 does not account for selection of the active population noted above, and thus actually estimates the mean total duration among the active population $\mathbb{E}[x|a]$, which we know to be greater than among the source population $\mathbb{E}[x]$ per Eq. (2).²

If total durations among the source population are exponentially distributed: $p(x) \sim \text{Exp}(\lambda)$, then $\mathbb{CV}[x] = 1$, and so the mean *censored* duration among the *active* population will directly equal the mean *total* duration among the *source* population (without doubling): $\mathbb{E}[z|a] = \mathbb{E}[x]$. In other words, under these conditions, the mean reported duration in a typical cross-sectional survey will be an unbiased estimator of the target $\mathbb{E}[x]$. Moreover, even the *distributions* will be identical: $p(z|a) = p(x) = \text{Exp}(\lambda)$, as we will show below. This is particularly notable since compartmental models usually assume exponentially distributed durations.

Complete distribution By combining Eqs. (1) and (3), we can derive the complete distribution of censored durations among the active population $p(z|a)$ from the distribution of total durations among the source population $p(x)$ [7]:

$$\begin{aligned} p(z|a) &= \int_0^\infty p(z|x) p(x|a) dx \\ &= \int_z^\infty \frac{1}{x} p(x|a) dx \\ &= \frac{1}{K} \int_z^\infty \frac{x}{x} p(x) dx \\ &= \frac{1}{K} (1 - F_x(z)) \end{aligned} \quad (4)$$

² Fazito et al. [4] partially acknowledge this issue by opting to use the median over the mean of $z|a$ where possible, since the median is less sensitive to outliers. However, as shown in Figures A.5 and A.6, doubling the median $\mathbb{M}[z|a]$ still overestimates $\mathbb{E}[x]$, just to a lesser degree than doubling the mean $\mathbb{E}[z|a]$, since the median will always be less than the mean for right-skewed distributions.

Interestingly, this is simply one minus the cumulative distribution of total durations F_x , normalized to have unit area under the curve (via K).³ For the exponential case, we have: $F_x = 1 - e^{-\lambda x}$, $K = \lambda^{-1}$, and thus $p(z|a) = \lambda e^{-\lambda x} = p(x)$.

More generally, $\text{CV}[z|a]$ is typically drawn towards 1 relative to $\text{CV}[x]$, attenuating skewness when $\text{CV}[x] > 1$, and amplifying skewness when $\text{CV}[x] < 1$ (Figures 2 and A.5). However, we believe that $\text{CV}[x] > 1$ is more likely than $\text{CV}[x] < 1$ for two reasons. First, heavy-tailed distributions having $\text{CV}[x] > 1$ arise naturally via selection effects in populations with heterogeneous rates [8]. Second, most parametric distributions with $\text{CV}[x] < 1$ have $p(x \rightarrow 0) \rightarrow 0$ (Figure A.5), which effectively implies that few people exit sex work shortly after starting, or that most people stay in sex work for a "minimum" period, which seems unrealistic. If $\text{CV}[x] > 1$ is indeed more likely, then the skewness of the observable duration distribution $p(z|a)$ will generally underestimate the skewness of the true duration distribution $p(x)$.

Sampling Biases If the survey undersamples individuals in the early years of sex work, this probability of being sampled will be a function of current duration (z) not total duration (x), so we denote it $p(s|z)$. Many parameterizations are possible; e.g. Eq. (7). The distribution of observed durations among survey participants is then updated from Eq. (4) simply as:

$$p(z|s) = \frac{1}{K} p(s|z) (1 - F_x(z)) \quad (5)$$

Estimating $\mathbb{E}[x]$ We conclude this section by revisiting the three methods considered by Fazito et al. [4] to estimate the "true" mean duration in sex work $\mathbb{E}[x]$. Figure A.6 plots: the typically unobserved mean and median of total durations (method 1), *double* the means and medians of censored durations (method 3), and the reciprocal proportion of censored durations less than 1 year (method 2), across a range of $\text{CV}[x]$ and distribution families. We see that at low $\text{CV}[x]$, nearly all methods converge on $\mathbb{E}[x]$. However, as $\text{CV}[x]$ increases, selection towards longer total durations among the active/sampled populations raises the means of the corresponding censored durations $\mathbb{E}[z|\cdot]$, according to Eq. (2). Median censored durations exhibit a similar trend, but lag below means (true for all right-skewed distributions) to a degree which depends on the family, making such medians less useful estimators of $\mathbb{E}[x]$. For all method 3 estimates, undersampling of early sex work (here assuming $\rho_s = 0.5$, $\tau_{95} = 1$ year) predictably biases estimates slightly higher; this bias is also larger for smaller $\mathbb{E}[x]$ (not shown). Finally, turnover-based estimates (method 2) tends to be most consistent, only slightly overestimating $\mathbb{E}[x]$ as $\text{CV}[x]$ increases; however, this method is especially sensitive to undersampling of early sex work, which biases estimates upwards.

Nevertheless, the best simple estimate of $\mathbb{E}[x]$ is likely $\mathbb{E}[z|s]$, assuming $\text{CV}[x] \approx 1$, or more generally:

$$\hat{\mathbb{E}}[x] = \frac{1}{2} \mathbb{E}[z|s] (1 + \text{CV}^2[x]) \quad (6)$$

³ The integral lower bound changes from 0 to z because $p(z|x) = \frac{1}{x}$ only when $x < z$ and $p(z|x) = 0$ otherwise, per Eq. (3).

4 Application

We highlight the utility of the above analytic relationships through a Bayesian model, which aims to infer the complete distribution of total durations among the source population $p(x)$ from individual-level and/or aggregate data on the distribution of censored durations among the sampled population $p(z|s)$. Specifically, we focus on the types of aggregate estimates reviewed by Fazito et al. [4]: distribution means $\hat{E}[z|s]$, and proportions of durations within a given interval $[a, b]$, denoted $\hat{P}[a < z < b|s]$, which can encompass proportions of responses falling within pre-specified ranges, as well as estimated medians, quartiles, etc.

4.1 Bayesian Model

The model assumes a parametric distribution for total durations among the source population $p(x)$. Specifically, we considered 6 distribution families: Exponential, Gamma, Weibull, Log-Normal, Scaled Beta, and Scaled Kumaraswamy, all truncated or scaled to 50 years maximum duration (Table A.1). For convenience, we denote the parameters of all families as: α, β (for Exponential: β only), so the interpretation of and priors for each parameter depend on the family.

We model the probability of being sampled for survey participation among the active population as increasing exponentially from $\rho_s \in [0.5, 1]$ initially to 95% by year $\tau_{95} \in [0, 3]$, and 100% asymptotically:⁴

$$p(s|z) = 1 + (\rho_s - 1)e^{\left(\frac{z}{\tau_{95}} \log\left(\frac{0.05}{1-\rho_s}\right)\right)} \quad (7)$$

We then define the distribution of censored durations among the sampled population $p(z|s; \alpha, \beta, \rho_s, \tau_{95})$ via Eq. (5). What remains is to specify likelihood functions for estimates $\hat{E}[z|s]$ and $\hat{P}[a < z < b|s]$ from empirical data, which are defined in Eqs. (A.1) and (A.2), respectively.

Under this model, we infer posterior distributions for $\alpha, \beta, \rho_s, \tau_{95}$ given the data. From α, β , we can then compute the distribution of total durations among the source population $p(x; \alpha, \beta)$, and any characteristics thereof, e.g. the mean $E[x; \alpha, \beta]$. We implemented and fit the model using Stan [10], retaining 1000 samples after 500 warm-up iterations from 7 chains with default random initialization and sampling algorithm [11]. We confirmed convergence visually through parameter trace plots (not shown).

4.2 Single Study

We first apply the model to aggregate data from a single study: a 2011 cross-sectional survey of 328 female sex workers in Eswatini [9]. The study reported crude and RDS-adjusted mean (95% CI) proportions of respondents who had been selling sex for 0–3, 3–6, 6–11, and 11+ years (Table A.2). We converted these 95% CI to effective sample sizes per stratum (N_i), for compatibility with Eq. (A.2).

⁴ Thus, if either $\rho_s = 1$ or $\tau_{95} = 0$, then sampling will be modelled as perfectly representative of the active population: $p(s|z) = 1$.

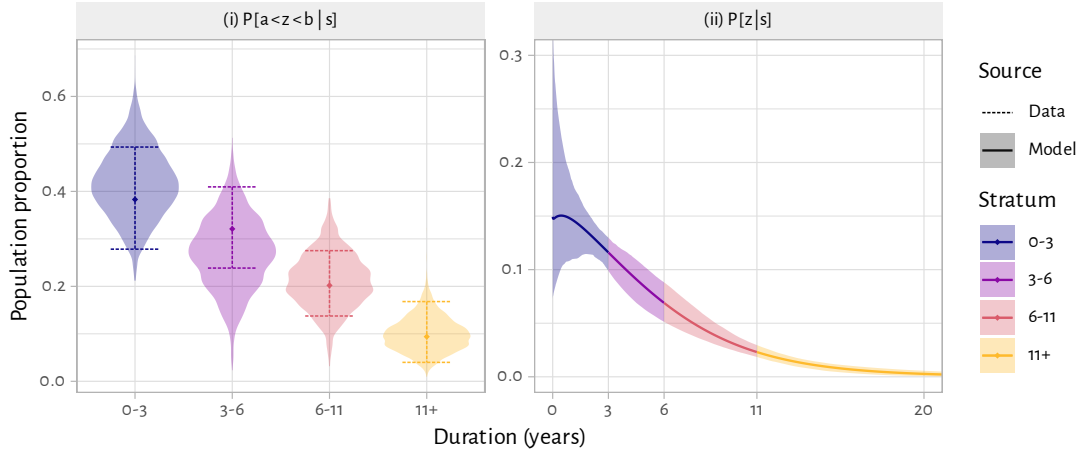


Figure 3: Posterior estimates from fitting the Bayesian model to RDS-adjusted aggregate censored duration data from [9]: (i) proportions of respondents with durations in pre-specified ranges; (ii) inferred duration density distribution $p(z|s)$

Shaded regions denote: (i) posterior of modelled proportions; (ii) posterior 95% CI of modelled duration distribution. Points and dashed line ranges denote reported 95% CI for proportion data.

Figure 3 illustrates (i) the posterior distributions of modelled proportions vs the RDS-adjusted data, and (ii) the corresponding estimated distribution $p(z|s)$ for the best-fitting family (Gamma). Figure A.7 further illustrates the posterior distributions of model parameters and summary statistics for $p(x)$ and $p(z|s)$ across all families considered. We find that the model can reproduce these data quite well. However, considerable uncertainty remains regarding the proportions of respondents selling sex for less than 1 year, mainly because the available data aggregate the first 0–3 years into one stratum. This issue also limits our ability to infer parameters of the sampling model: ρ_s, τ_{95} , whose posterior distributions are effectively indistinguishable from their priors.

The posterior distribution of $\mathbb{E}[z|s]$ was reasonably precise and consistent across the families considered: approximately 5.0 (95% CI: 4.4, 5.8), and likewise for $\mathbb{CV}[z|s]$: 0.9 (0.7, 1.2). By contrast, estimates of $\mathbb{E}[x]$ and $\mathbb{CV}[x]$ were more varied, with 95% CI extremes spanning (2.0, 8.8) and (0.4, 1.9), respectively, and best-fitting $\mathbb{E}[x] = 5.7$, $\mathbb{CV}[x] = 0.9$ under the Gamma family.

4.3 Evidence Synthesis

Noting the limitations of previously considered methods to estimate $\mathbb{E}[x]$, we additionally applied the model to re-synthesize duration data reviewed by Fazito et al. [4].

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Appendix

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Appendix A

Supplementary Results

A.1 Parametric equations

Distribution families We considered 6 distribution families for $p(x)$ in our analysis (Table A.1).

Table A.1: Distribution families considered, using common parameter notation (α, β)

Family	Definition $p(x)$	α log-prior [*]	β log-prior [*]
Exponential [†]	$\frac{1}{\alpha} e^{-\frac{x}{\alpha}}$	–	[–1, 3]
Gamma [†]	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$	[–2, 2]	[–3, 3]
Weibull [†]	$\frac{\alpha}{\beta} \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-(x/\beta)^\alpha}$	[–1, 2]	[–1, 2]
Log-Normal [†]	$\frac{1}{x \beta \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{\log x - \alpha}{\beta}\right)^2}$	[–9, 1]	[–2, 3]
Scaled Beta [‡]	$\frac{x^{\alpha-1} (1 - x^\alpha)^{\beta-1}}{B(\alpha, \beta)}$	[–3, 3]	[0, 3]
Scaled Kumaraswamy [‡]	$\alpha \beta x^{\alpha-1} (1 - x^\alpha)^{\beta-1}$	[–2, 3]	[0, 3]

^{*} Priors for α, β in the applied model were defined as uniform on the log-scale; e.g. [–1, 3] corresponds to $[e^{-1}, e^3] \approx [0.37, 20]$.

[†] Distributions spanning $[0, \infty)$ were truncated to $[0, 50]$ years. [‡] Distributions spanning $[0, 1]$ were scaled to span $[0, 50]$ years.

Likelihoods functions Under the model for $p(z|s)$, we define the likelihoods for empirical estimates of the mean $\hat{\mathbb{E}}_i[z|s]$ and proportions $\hat{\mathbb{P}}_i[a < z < b|s]$ from a given survey with sample size N_i as, respectively: the Gamma-equivalent of Normal sampling error for the mean (since $\hat{\mathbb{E}} > 0$), and the Beta-approximation of the Binomial distribution (for differentiability):

$$\hat{\mathbb{E}}_i[z|s; N_i] \sim \text{Gamma}(\alpha = N_i / \mathbb{CV}_i^2[z|s], \beta = N_i / \mathbb{CV}_i^2[z|s] / \mathbb{E}[z|s]) \quad (\text{A.1})$$

$$\hat{\mathbb{P}}_i[a < z < b|s; N_i] \sim \text{Beta}(\alpha = N_i \mathbb{P}_i[a < z < b|s], \beta = N_i (1 - \mathbb{P}_i[a < z < b|s])) \quad (\text{A.2})$$

To validate these likelihood functions, we simulated 10,000 stochastic samples of z with sample sizes $N_i = \{30, 100, 300, 1000\}$ given known $p(z|s)$ under $\mathbb{E}[x] = 5$, $\mathbb{CV}[x] = \{0.5, 1, 1.5\}$, and all 6 distribution families. We then compared the distribution of stochastic estimates of $\hat{\mathbb{E}}[z|s; N_i]$ and $\hat{\mathbb{P}}[a < z < b|s; N_i]$ from each sample with the analytic likelihood functions: Eqs. (A.1) and (A.2). For proportions, we considered $\mathbb{P}[0 < z < 1]$, $\mathbb{P}[1 < z < 5]$, and $\mathbb{P}[5 < z < 9]$, arbitrarily.

Figures A.1–A.4 illustrate these comparisons, where we observe good agreement between the stochastic and analytic distributions across all conditions. Note that stochastic distributions for $\hat{\mathbb{P}}[a < z < b|s; N_i]$ are discrete (more visible at low N_i) because there are a finite number simulated respondents whose durations can fall in the specified interval $[a, b]$.

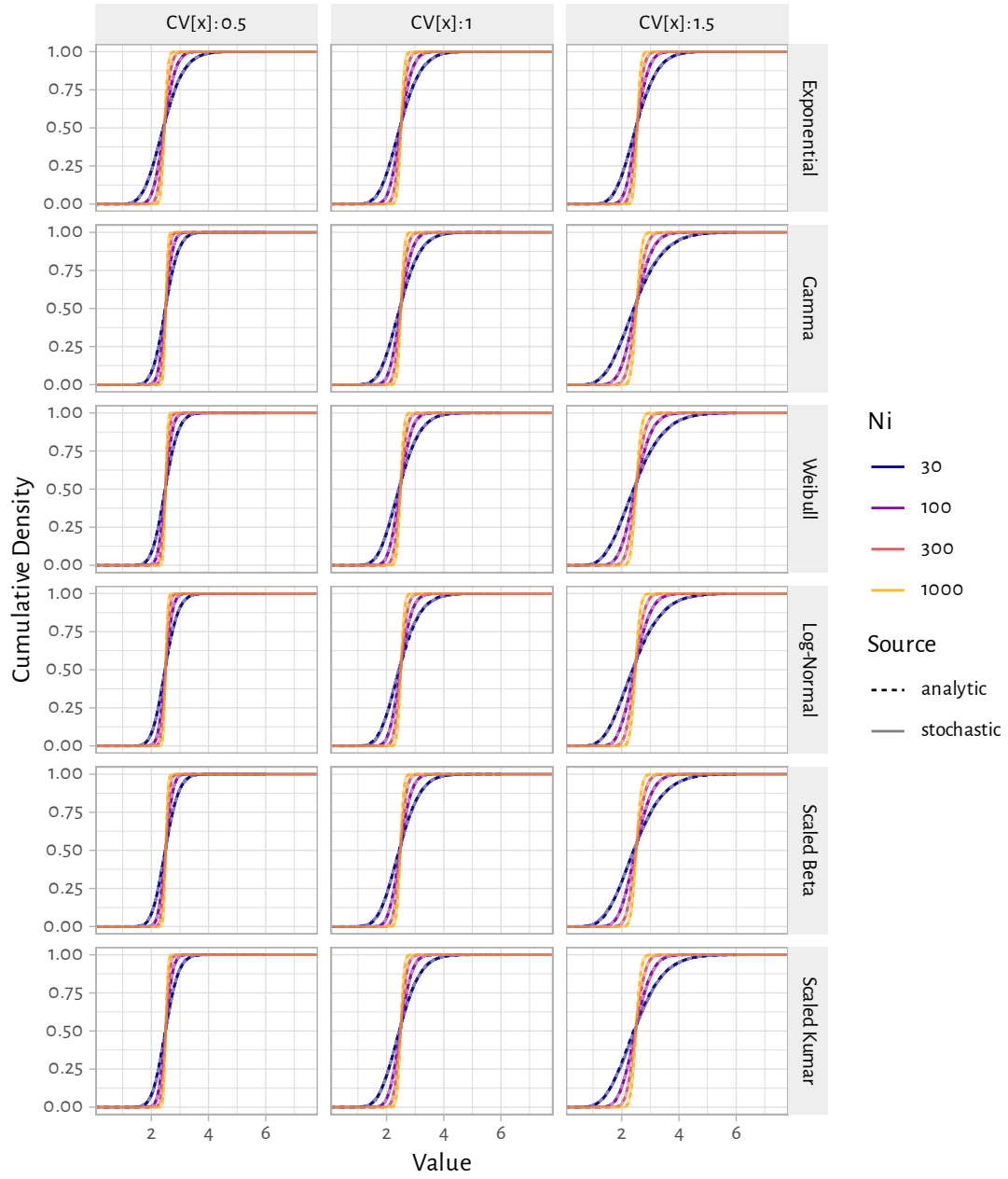


Figure A.1: Comparison of analytic likelihood functions and distribution of 10,000 stochastic estimates of $\mathbb{E}[\mathbf{z}|\mathbf{s}]$, for different simulated survey sizes N_i and latent distribution families for $p(x)$

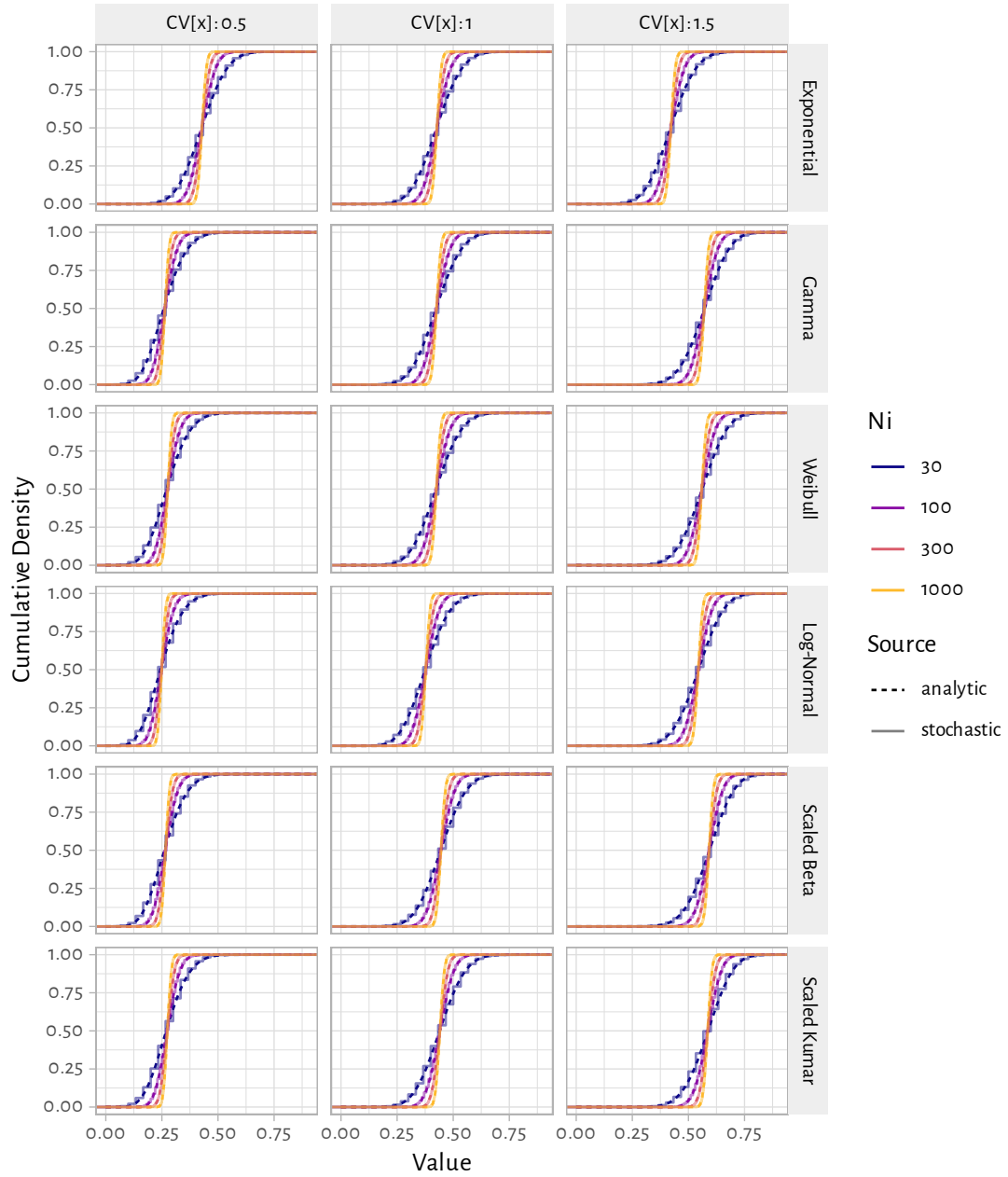


Figure A.2: Comparison of analytic likelihood functions and distribution of 10,000 stochastic estimates of $\mathbb{P}[\mathbf{0} < \mathbf{z} < \mathbf{1} | \mathbf{s}]$, for different simulated survey sizes N_i and latent distribution families for $p(x)$

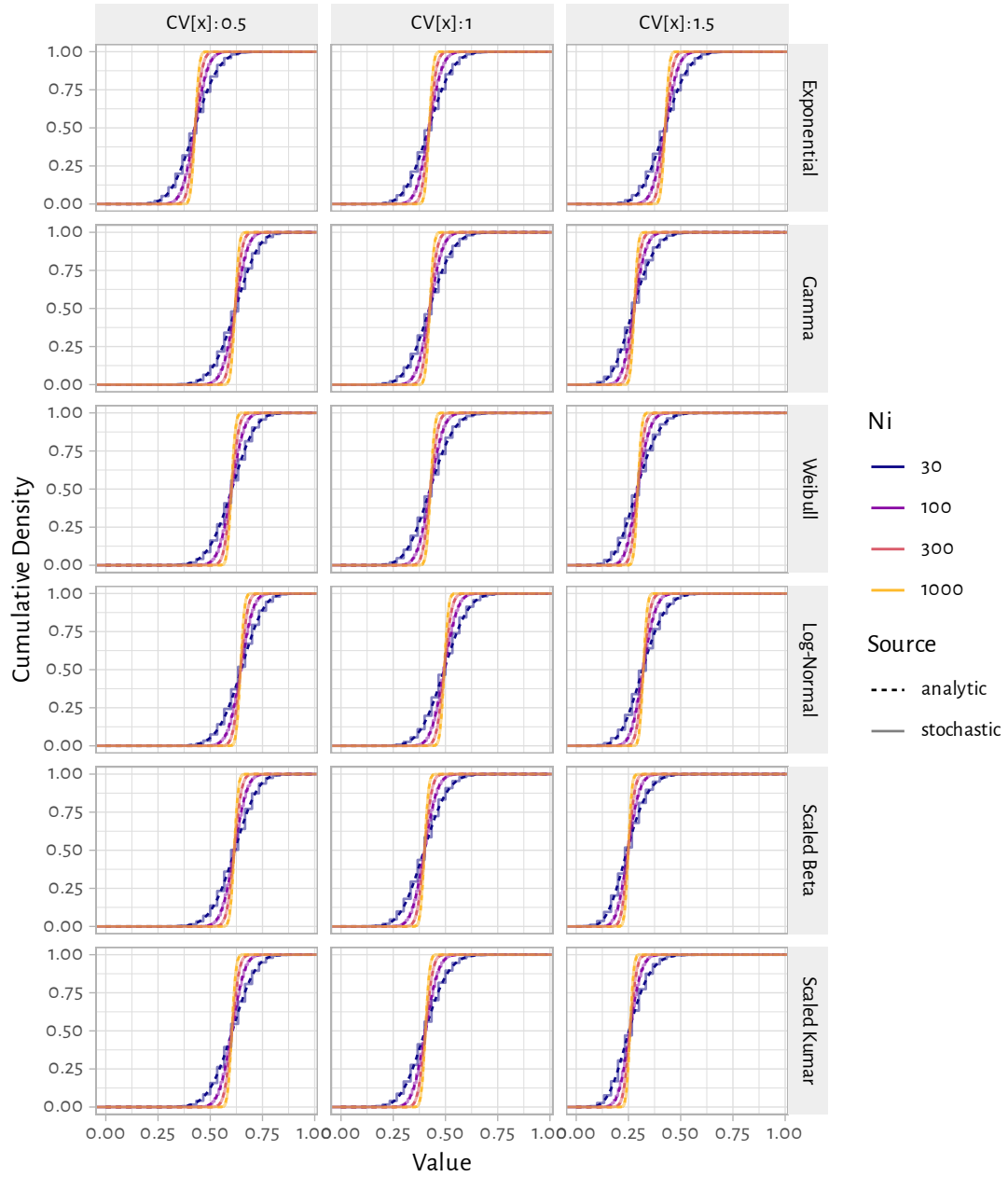


Figure A.3: Comparison of analytic likelihood functions and distribution of 10,000 stochastic estimates of $\mathbb{P}[1 < \mathbf{z} < 5|\mathbf{s}]$, for different simulated survey sizes N_i and latent distribution families for $p(\mathbf{x})$

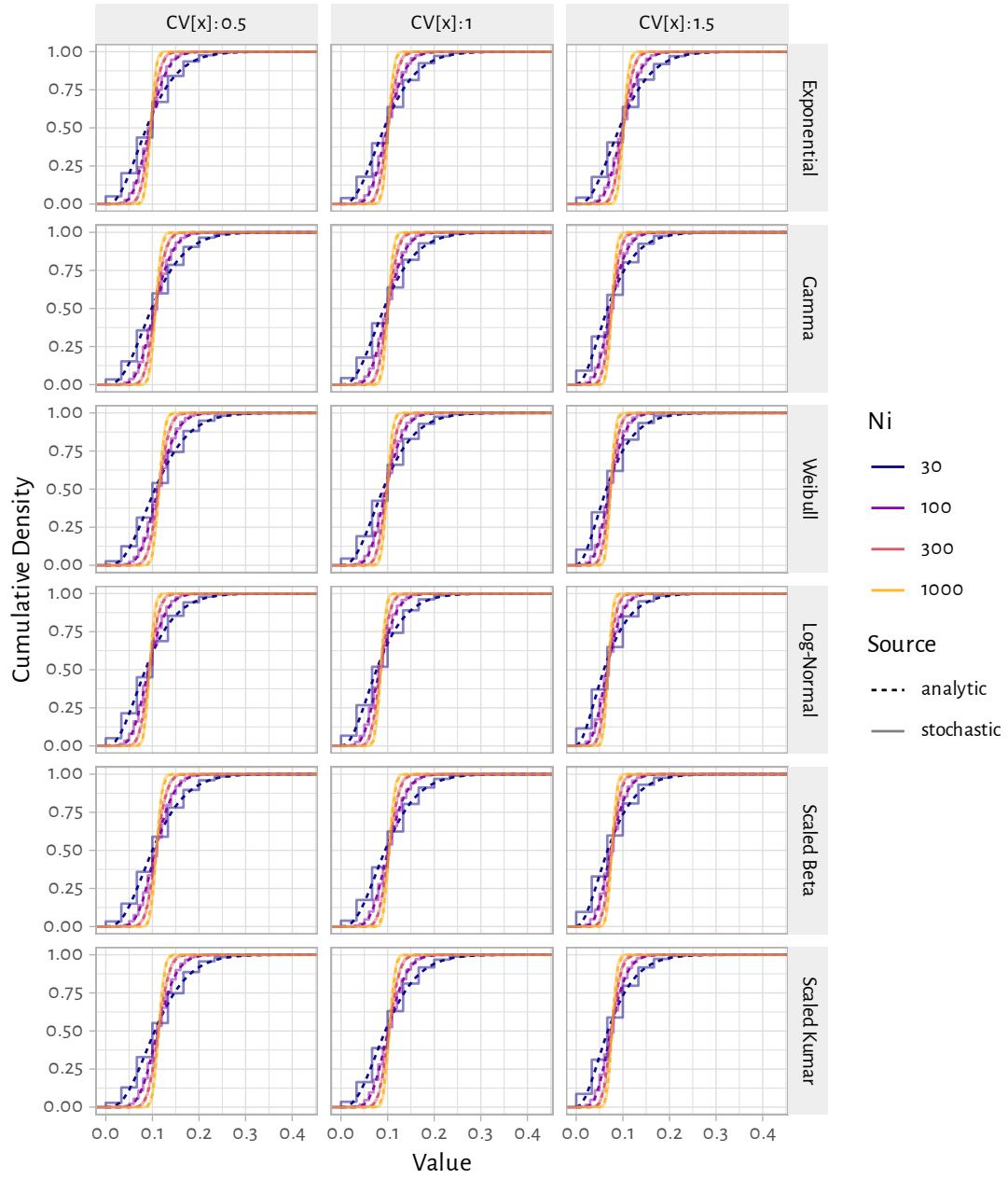


Figure A.4: Comparison of analytic likelihood functions and distribution of 10,000 stochastic estimates of $\mathbb{P}[5 < \mathbf{z} < 9|\mathbf{s}]$, for different simulated survey sizes N_i and latent distribution families for $p(\mathbf{x})$

A.2 Illustrative analytic results

Figure A.5 repeats Figure 2 for 5 distribution families considered (Table A.1) — i.e. showing distributions of total (x) and censored (z) durations among the source, active, and sampled populations for each family, all having $\mathbb{E}[x] = 5$ and $\text{CV}[x] = \{0.5, 1, 1.5\}$. We omit the Exponential family as it is a special case of both Gamma and Weibull with $\text{CV}[x] = 1$. Figure A.6 further illustrates performance of different $\mathbb{E}[x]$ estimators considered in [2] across a range of $\text{CV}[x]$ and the same 5 distribution families.

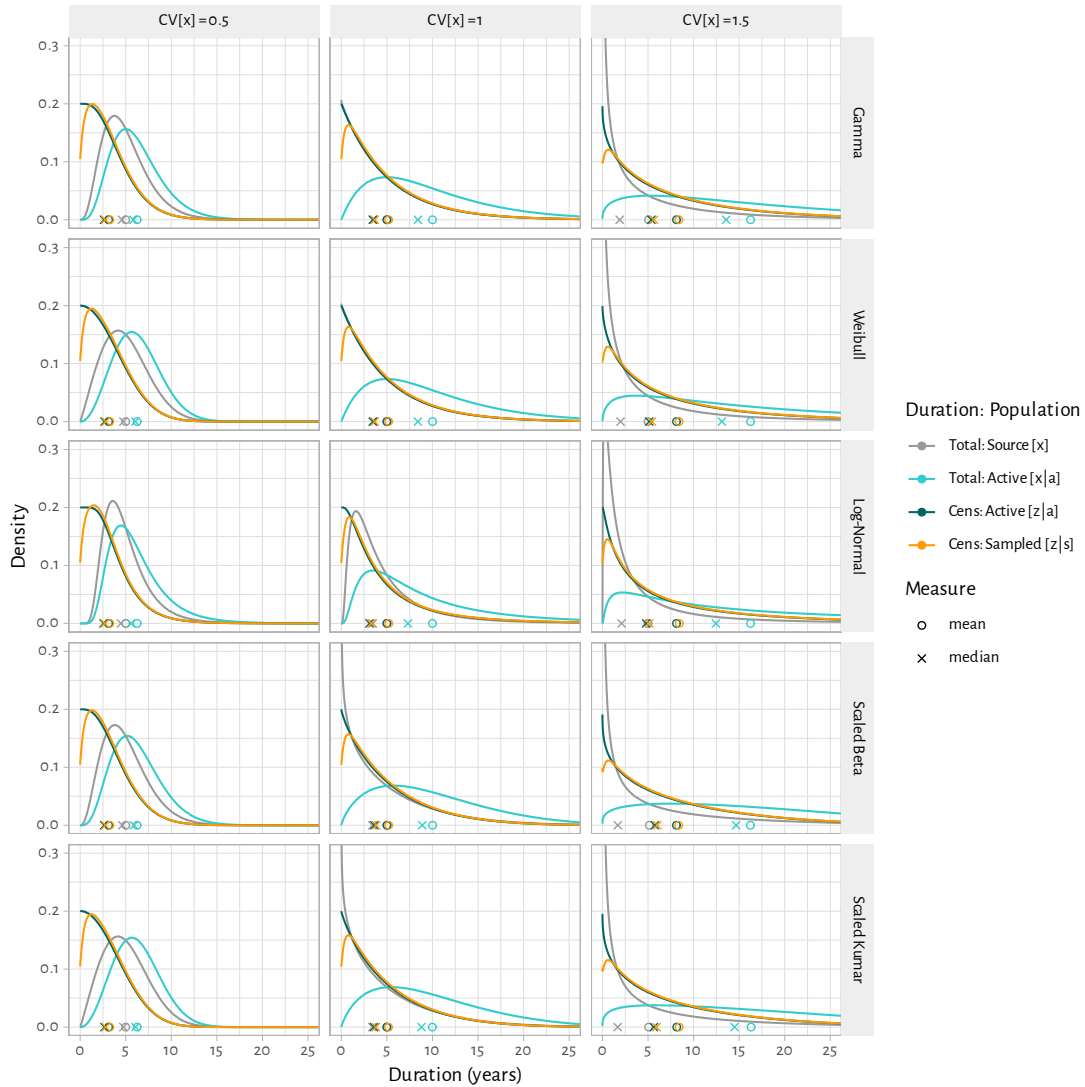


Figure A.5: Illustrative distributions of total and censored durations among the source, active, and sampled populations for different x distribution families (columns) and 3 coefficients of variation $\text{CV}[x]$ (rows)

Note: true $\mathbb{E}[x] = 5$; probability of being sampled modelled per Eq. (7) with $\rho_s = 0.5$, $\tau_{95} = 1$.

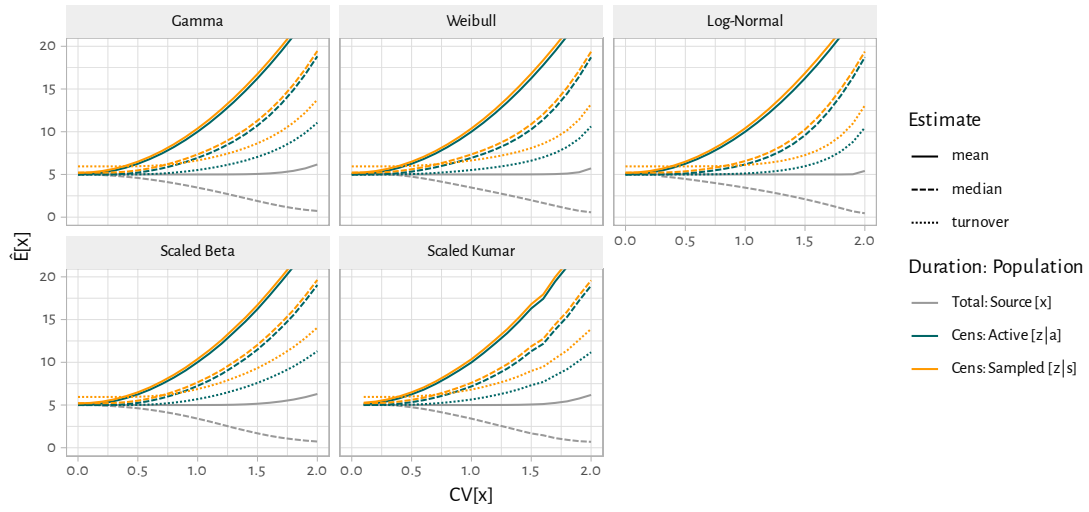


Figure A.6: Performance of $E[x]$ estimators considered in [2] across a range of $CV[x]$ and distribution families

Note: true $E[x] = 5$; probability of being sampled modelled per Eq. (7) with $\rho_s = 0.5$, $\tau_{95} = 1$; means and medians for censored data (z) are doubled per [2].

A.3 Application

Eswatini data Table A.2 summarizes the RDS-adjusted sex work duration data reported in [1], while Figure A.7 illustrates the posterior distributions inferred by fitting our Bayesian model to those data.

Table A.2: RDS-adjusted proportions of durations selling sex among Swati female sex workers in [1] (N = 313)

Duration [†] (years)	Proportion (%)			N_i Eff. [‡]
	Crude	RDS	(95% CI)	
0–3	28.8	38.3	(27.5, 49.1)	77
3–6	31.3	32.1	(23.6, 40.7)	113
6–11	26.2	20.2	(13.2, 27.1)	129
11+	13.7	9.4	(4.4, 14.4)	135

[†] Reported strata in [1] were: "under 2, 3–5, 6–10, 11 or more", which we corrected here to avoid implied gaps between strata, [‡] Effective sample size per strata calculated to match RDS-adjusted 95% CI.

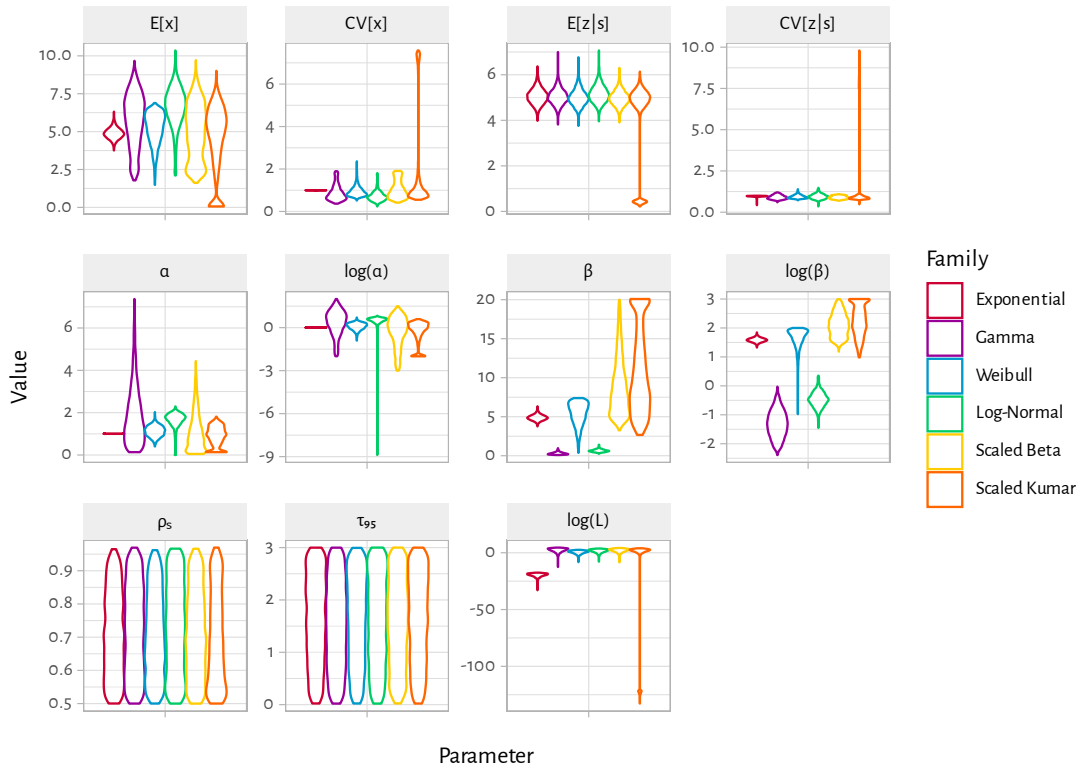


Figure A.7: Posterior distribution of mean & CV total/censored distributions among source/sampled populations: $E[x]$, $CV[x]$ / $E[z|s]$, $CV[z|s]$; model parameters: α , β , ρ_s , τ_{95} ; and log-likelihood: $\log(L)$ across 6 distribution families considered when fitting to the RDS-adjusted Eswatini data (Table A.2)

Global data [2]